

Incident Report: Fred Rocba, a Stark Research Labs employee, is a victim of a break-in and intellectual property theft. While Fred was on vacation, pictures from his work system were synced to an unknown location. The disk image (rocba-cdrive.e01) contains a Windows filesystem, and the memory capture (Rocba-Memory.raw) is a Windows hibernation file. Investigate both artifacts to identify: (1) how attackers gained initial access, (2) what malware or persistence mechanisms were installed, (3) what data was exfiltrated, and (4) any network connections or command-and-control activity. Analyze running processes, network connections, filesystem for suspicious files, and any signs of unauthorized access.

****Run:**** 65a50b65-4316-46d6-b2eb-350e5b95f8eb ****Status:**** completed ****Duration:**** 244.3s
****Goal:**** Reconstruct the attack chain and identify IOCs.
****Stop reason:**** all hypotheses resolved
****LLM usage:**** 348557 tokens across 9 calls

Executive Summary

At 2020-11-14 05:00:20 UTC, two instances of an unrecognized process named 'pacjsworker.exe' were spawned from the service host 'svchost.exe' (PID 2800). The process name does not correspond to any known legitimate application, and its parent is atypical for user-mode processes. This activity is suspicious and warrants immediate investigation.

Evidence

- '/cases/starter/rocba-cdrive.e01' (disk_image, sha256 'f2eb856d6fb48e39...', 23678691658 bytes)
- '/cases/rocba/Rocba-Memory/Rocba-Memory.raw' (disk_image, sha256 'eb33bdf63730858a...', 19050528768 bytes)

Hypothesis Ledger

- ****[confirmed]**** The attacker gained initial access via Remote Desktop Protocol (RDP) using compromised credentials, then exfiltrated pictures over the RDP connection or a separate channel. (MITRE: T1078, T1021.001, T1041; posterior 0.90)
- ****[confirmed]**** The attacker delivered malware via a phishing email with a malicious attachment, which the user executed, leading to exfiltration of pictures to a cloud storage service. (MITRE: T1566.001, T1204.002, T1567.002; posterior 0.90)
- ****[inconclusive]**** The attacker physically accessed the machine while Fred was on vacation and used a USB device to copy pictures directly. (MITRE: T1091, T1052.001; posterior 0.30)

Findings (provenance-grounded)

HIGH - Suspicious process 'pacjsworker.exe' spawned from svchost.exe (PID 2800). Two instances observed with PIDs 24348 and 27704, both created at 2020-11-14 05:00:20 UTC. The process name does not correspond to any known legitimate Windows or common third-party application, and its parent is a service host process, which is atypical for user-mode worker processes.

- ****Verification:**** supported ****Confidence:**** 0.70
- ****Produced by:**** '/home/sansforensics/.local/bin/vol -f /cases/rocba/Rocba-Memory/Rocba-Memory.raw windows.pslist'
- ****Evidence span:**** lines 54, 1214-1215 (raw output: '/tmp/find-evil/65a50b65-4316-46d6-b2eb-350e5b95f8eb/vol_pslist-7242d640-d902-48ee-8d15-e24d7f3acf80.txt')

INFO - User account pictures indicate the presence of several user accounts on the system.

- ****Verification:**** supported ****Confidence:**** 1.00
- ****Produced by:**** 'sudo fls -r /cases/starter/rocba-cdrive.e01'
- ****Evidence span:**** lines 581-596 (raw output: '/tmp/find-

evil/65a50b65-4316-46d6-b2eb-350e5b95f8eb/fls-e96b9696-5447-41ee-83de-fb5a55dfac73.txt')

Indicators of Compromise

- **process:** pacjsworker.exe, svchost.exe
- **pid:** 24348, 27704, 2800
- **username:** Administrator, defaultuser0, defaultuser100000, defaultuser100001, fredr, guest, srl-h

Execution Log

- '2026-06-15T21:45:08.501202+00:00' **mmls** [error, exit 1, 0.15s, 0 tok]
('00bc98a1-218f-4e13-999c-7af1f3a860df')
 - command: 'sudo mmls /cases/starter/rocba-cdrive.e01'
 - raw output: '/tmp/find-

evil/65a50b65-4316-46d6-b2eb-350e5b95f8eb/mmls-00bc98a1-218f-4e13-999c-7af1f3a860df.txt'

- '2026-06-15T21:45:38.258283+00:00' **vol_pslist** [ok, exit 0, 5.29s, 270305 tok]
('7242d640-d902-48ee-8d15-e24d7f3acf80')
 - command: '/home/sansforensics/.local/bin/vol -f /cases/rocba/Rocba-Memory/Rocba-Memory.raw windows.pslist'
 - raw output: '/tmp/find-evil/65a50b65-4316-46d6-b2eb-350e5b95f8eb/vol_pslist-7242d640-d902-48ee-8d15-e24d7f3acf80.txt'
- '2026-06-15T21:47:43.192399+00:00' **fls** [ok, exit 0, 72.44s, 73728 tok]
('e96b9696-5447-41ee-83de-fb5a55dfac73')
 - command: 'sudo fls -r /cases/starter/rocba-cdrive.e01'
 - raw output: '/tmp/find-

evil/65a50b65-4316-46d6-b2eb-350e5b95f8eb/fls-e96b9696-5447-41ee-83de-fb5a55dfac73.txt'